

Finite State Machines?

Your compiler wants in!

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IMPLICIT STATE

THE VALLEY OF PROGRAMMER DEATH

ADTs

IDRIS

TYPE-SAFE SHORES

TYPE-LEVEL MAPS

INDEXED MONADS

TYPE CLASSES

SIDE EFFECTS

MTL STYLE

@owickstrom

State

Stateful Programs

- The program *remembers* previous events
- It may transition to another state based on its current state

- The program does not explicitly define the set of legal states
- State is scattered across many mutable variables
- Hard to follow and to ensure the integrity of state transitions
- Runtime checks “just to be sure”

Making State Explicit

- Instead, we can make states *explicit*
- It is clearer how we transition between states
- Make stateful programming less error-prone

Finite-State Machines

- We model a program as an abstract *machine*
- The machine has a finite set of *states*
- The machine is in one state at a time
- *Events* trigger state transitions
- From each state, there's a set of legal transitions, expressed as associations from events to other states

$$\textit{State}(S) \times \textit{Event}(E) \rightarrow \textit{Actions}(A), \textit{State}(S')$$

If we are in state S and the event E occurs, we should perform the actions A and make a transition to the state S' .

— Erlang FSM Design Principles ¹

¹ http://erlang.org/documentation/doc-4.8.2/doc/design_principles/fsm.html

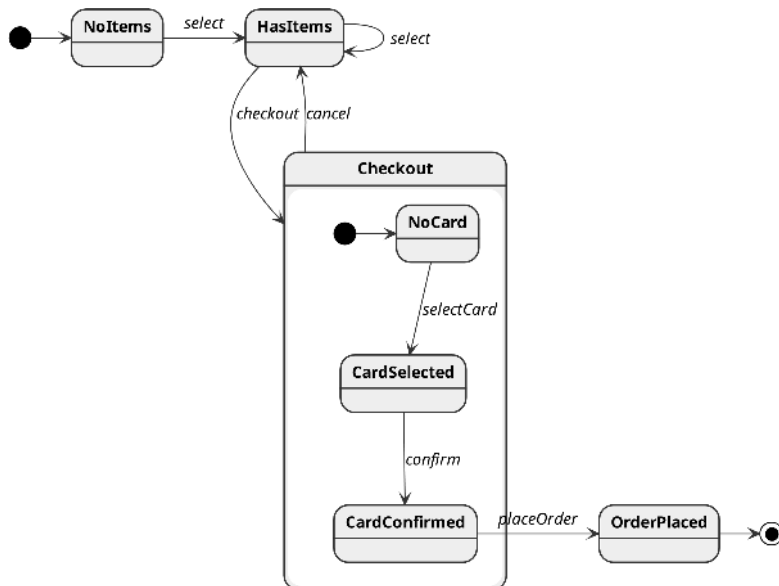
- Not strictly Mealy or Moore machines
- No hierarchical machines
- No guards in our models
- No UML statecharts



- We model the set of legal states as a data type
- Each state has its own value constructor
- You can do this in most programming languages
- We'll use Haskell to start with

Encoding with Algebraic Data Types

Example: Checkout Flow



States as an ADT

```
data CheckoutState
  = NoItems
  | HasItems (NonEmpty CartItem)
  | NoCard (NonEmpty CartItem)
  | CardSelected (NonEmpty CartItem)
                    Card
  | CardConfirmed (NonEmpty CartItem)
                    Card
  | OrderPlaced
deriving (Show, Eq)
```

Events as an ADT

```
data CheckoutEvent
  = Select CartItem
  | Checkout
  | SelectCard Card
  | Confirm
  | PlaceOrder
  | Cancel
deriving (Show, Eq)
```

FSM Type

```
type FSM s e =  
  s -> e -> s
```

```
checkout :: FSM CheckoutState CheckoutEvent
```


State Machine with IO

```
type ImpureFSM s e =  
  s -> e -> IO s
```

Checkout using ImpureFSM

```
checkoutImpure :: ImpureFSM CheckoutState CheckoutEvent
```

Checkout using ImpureFSM (cont.)

```
checkoutImpure NoItems (Select item) =  
  return (HasItems (item :| []))
```

```
checkoutImpure (HasItems items) (Select item) =  
  return (HasItems (item <| items))
```

...

Checkout using ImpureFSM (cont.)

...

```
checkoutImpure (CardConfirmed items card) PlaceOrder = do  
  PaymentProvider.chargeCard card (calculatePrice items)  
  return OrderPlaced
```

Impure Runner

```
runImpure :: ImpureFSM s e -> s -> [e] -> IO s  
runImpure = foldM
```

Logging FSM

```
withLogging ::  
    (Show s, Show e)  
=> ImpureFSM s e  
-> ImpureFSM s e  
withLogging fsm s e = do  
    s' <- fsm s e  
    liftIO $  
        printf "- %s x %s → %s\n" (show s) (show e) (show s')  
    return s'
```

Impure Runner Example

```
runImpure
  (withLogging checkoutImpure)
  NoItems
  [ Select "food"
    , Select "fish"
    , Checkout
    , SelectCard "visa"
    , Confirm
    , PlaceOrder
  ]
```

Impure Runner Example Output

```
- NoItems × Select "food" → HasItems ("food" :| [])  
- HasItems ("food" :| []) × Select "fish" → HasItems ("fish" :| ["food"])  
- HasItems ("fish" :| ["food"]) × Checkout → NoCard ("fish" :| ["food"])  
- NoCard ("fish" :| ["food"]) × SelectCard "visa" → CardSelected ("fish" :| ["food"]) "visa"  
- CardSelected ("fish" :| ["food"]) "visa" × Confirm → CardConfirmed ("fish" :| ["food"]) "visa"  
Charging $666  
- CardConfirmed ("fish" :| ["food"]) "visa" × PlaceOrder → OrderPlaced
```


- We have explicit states using data types
- Standardized way of running state machine programs
 - It's simple to add logging, metrics
 - Instead of a list of events, we could use conduit² or pipes³
- We still have IO coupled with transitions (harder to test)
- Legal state transitions are not enforced

² <https://hackage.haskell.org/package/conduit>

³ <https://hackage.haskell.org/package/pipes>

MTL Style and Associated Types

MTL Style with an Associated Type

- We will write our state machines in “MTL style”
- Some extra conventions for state machines
- With MTL style, we can:
 - combine with monad transformers (error handling, logging, etc)
 - build higher-level machines out of lower-level machines

Typeclass and Abstract Program

- A typeclass encodes the state machine transitions
- Events are represented as typeclass methods
- The current state is passed as a *value*
- The state transitioned to is returned as a value
- The state type is *abstract* using an associated type alias
- We write a program depending on the typeclass
- The typeclass and the program together form the state machine

- An instance is required to run the state machine program
- The instance performs the state transition side-effects
- The instance chooses the concrete data type
- We can write test instances without side-effects

States as Empty Types

data NoItems

data HasItems

data NoCard

data CardSelected

data CardConfirmed

data OrderPlaced

State Machine with Class

```
class Checkout m where
  type State m :: * -> *

  ...
```

The `initial` method gives us our starting state:

```
initial :: m (State m NoItems)
```


Some events transition from exactly one state to another:

```
confirm ::  
  State m CardSelected -> m (State m CardConfirmed)
```

The Select Event

- Some events are accepted from many states
- Both NoItems and HasItems accept the select event
- We could use Either

Selection States

```
data SelectState m
  = NoItemsSelect (State m NoItems)
  | HasItemsSelect (State m HasItems)
```

Signature of select

```
select ::  
    SelectState m  
-> CartItem  
-> m (State m HasItems)
```

The Cancel Event

- There are *three* states accepting cancel
- Either would not work, only handles *two*
- Again, we create a datatype:

```
data CancelState m
  = NoCardCancel (State m NoCard)
  | CardSelectedCancel (State m CardSelected)
  | CardConfirmedCancel (State m CardConfirmed)
```

- And the signature of cancel is:

```
cancel :: CancelState m -> m (State m HasItems)
```

The Complete Typeclass

```
class Checkout {
```

```
  type State =
```

```
    initial: State
```

```
    select: State => State
```

```
    selectCard: State => State
```

```
    -> Checkout.State
```

```
    -> Checkout.State
```

```
    checkout: State => State
```

```
    selectCard: State => State
```

```
    State
```

```
    confirm: State => State
```

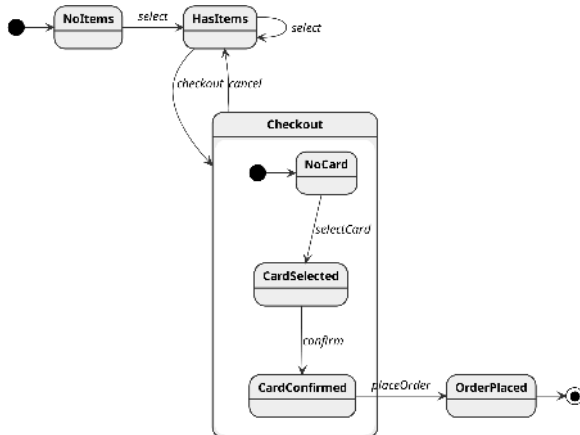
```
    State
```

```
    placeOrder: State => State
```

```
    State
```

```
    cancel: State => State
```

```
  end ::
```



A State Machine Program

```
fillCart ::  
    (Checkout m, MonadIO m)  
=> State m NoItems  
-> m (State m HasItems)  
fillCart noItems = do  
    first <- prompt "First item:"  
    select (NoItemsSelect noItems) first >>= selectMoreItems
```

A State Machine Program (cont.)

```
selectMoreItems ::  
    (Checkout m, MonadIO m)  
=> State m HasItems  
-> m (State m HasItems)  
selectMoreItems s = do  
    more <- confirmPrompt "More items?"  
    if more  
        then prompt "Next item:" >=>  
            select (HasItemsSelect s) >=>  
            selectMoreItems  
        else return s
```


A State Machine Program (cont.)

```
startCheckout ::  
    (Checkout m, MonadIO m)  
=> State m HasItems  
-> m (State m OrderPlaced)  
startCheckout hasItems = do  
    noCard <- checkout hasItems  
    card <- prompt "Card:"  
    cardSelected <- selectCard noCard card  
    useCard <-  
        confirmPrompt ("Confirm use of '" <> card <> "'?")  
    if useCard  
    then confirm cardSelected >>= placeOrder  
    else cancel (CardSelectedCancel cardSelected) >>=  
        selectMoreItems >>=  
            startCheckout
```

A State Machine Program (cont.)

```
checkoutProgram ::  
    (Checkout m, MonadIO m)  
    => m OrderId  
checkoutProgram =  
    initial >>= fillCart >>= startCheckout >>= end
```

The Abstract Part

- We only depend on the Checkout typeclass⁴
- Together with the typeclass, checkoutProgram forms the state machine

⁴ We do use MonadIO to drive the program, but that could be extracted.

A Checkout Instance

- We need an instance of the Checkout class
- It will decide the concrete State type
- The instance will perform the effects at state transitions
- We'll use it to run our checkoutProgram

Concrete State Data Type

```
data CheckoutState s where
  NoItems :: CheckoutState NoItems

  HasItems :: NonEmpty CartItem -> CheckoutState HasItems

  NoCard :: NonEmpty CartItem -> CheckoutState NoCard

  CardSelected
    :: NonEmpty CartItem
    -> Card
    -> CheckoutState CardSelected

  CardConfirmed
    :: NonEmpty CartItem
    -> Card
    -> CheckoutState CardConfirmed

  OrderPlaced :: OrderId -> CheckoutState OrderPlaced
```

CheckoutT

```
newtype CheckoutT m a = CheckoutT
  { runCheckoutT :: m a
  } deriving ( Monad
               , Functor
               , Applicative
               , MonadIO
               )
```

Checkout Instance

```
instance (MonadIO m) => Checkout (CheckoutT m) where
  type State (CheckoutT m) = CheckoutState

  ...
```

Initial State

...

```
initial = return NoItems
```

...

Select

...

```
select state item =
```

```
  case state of
```

```
    NoItemsSelect NoItems ->
```

```
      return (HasItems (item :| []))
```

```
    HasItemsSelect (HasItems items) ->
```

```
      return (HasItems (item <| items))
```

...

...

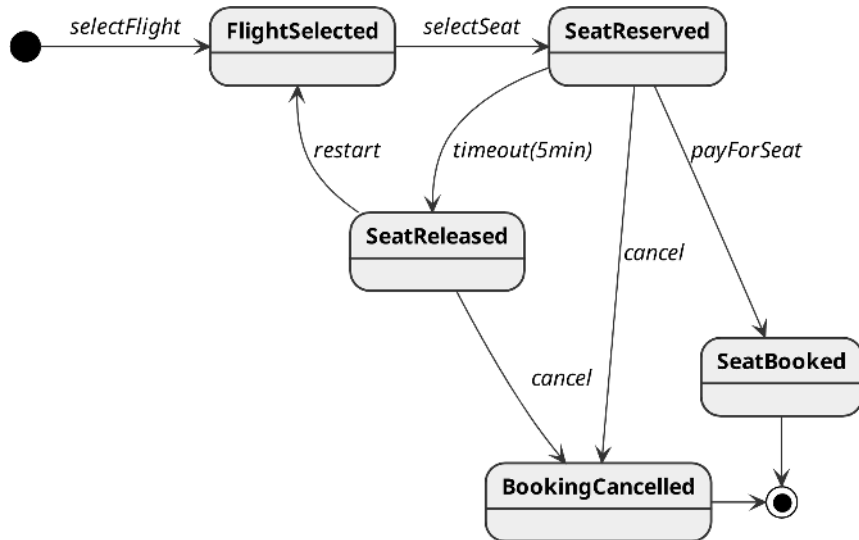
```
placeOrder (CardConfirmed items card) = do
  orderId <- newOrderId
  let price = calculatePrice items
  PaymentProvider.chargeCard card price
  return (OrderPlaced orderId)
```

Putting it all together

```
example :: IO ()  
example = do  
    orderId <- runCheckoutT checkoutProgram  
    T.putStrLn ("Completed with order ID: " <> orderId)
```

Timeouts

State Machine with Timeout



Flight Booking States

data FlightSelected

data SeatReserved

data SeatBooked

data BookingCancelled

data SeatReleased

Cancel and End States

```
data CancelState m
  = SeatReleasedCancel (State m SeatReleased)
  | SeatReservedCancel (State m SeatReserved)

data EndState m
  = CancelledEnd (State m BookingCancelled)
  | BookedEnd (State m SeatBooked)
```

Flight Booking Events and Transitions

```
class FlightBooking m where
  type State m :: * -> *
  selectFlight :: FlightNumber -> m (State m FlightSelected)
  selectSeat ::
    State m FlightSelected
    -> Seat
    -> m (State m SeatReserved)
  payForSeat ::
    State m SeatReserved -> m (State m SeatBooked)
  timeout ::
    State m SeatReserved -> m (State m SeatReleased)
  restart ::
    State m SeatReleased -> m (State m FlightSelected)
  cancel :: CancelState m -> m (State m BookingCancelled)
  end :: EndState m -> m ()
```


Flight Booking Program

```
bookFlight ::  
    (MonadBaseControl IO m, Monad m, FlightBooking m)  
    => m ()  
bookFlight =  
    prompt "Flight number:"  
    >>= selectFlight  
    >>= withFlightSelected
```

Flight Booking Timeout

```
withFlightSelected ::  
    (MonadBaseControl IO m, Monad m, FlightBooking m)  
=> State m FlightSelected -> m ()  
withFlightSelected flightSelected = do  
    seatReserved <- prompt "Seat:" >=> selectSeat flightSelected  
    answer <- Timeout.timeout (minutes 5) (confirmPrompt "Buy?")  
    case answer of  
  
        ...
```

Flight Booking Timeout (cont.)

...

```
Just True -> do
  seatBooked <- payForSeat seatReserved
  log "Seat booked!"
  end (BookedEnd seatBooked)
```

...

Flight Booking Timeout (cont.)

...

```
Just False -> do
  cancelled <- cancel (SeatReservedCancel seatReserved)
  log "Your booking has been cancelled."
  end (CancelledEnd cancelled)
```

...

Flight Booking Timeout (cont.)

...

```
Nothing -> do
  released <- timeout seatReserved
  wantsRestart <- confirmPrompt "Restart?"
  if wantsRestart
    then restart released >>= withFlightSelected
    else do
      cancelled <- cancel (SeatReleasedCancel released)
      end (CancelledEnd cancelled)
```

- We've modeled state machines using:
 - Type classes/MTL style
 - Associated types for states
 - Explicit state values
 - "Abstract" program
 - Instances for side-effects
- Stricter than ADT-based version
- Not necessarily safe
 - State values can be *reused* and *discarded*
 - Side-effects can be reperformed *illegally*
 - Nothing enforcing transition to a terminal state

Reusing State Values

```
payTwice seatReserved = do
  _ <- payForSeat seatReserved

  seatBooked <- payForSeat seatReserved
  log "You've payed twice, LOL."

end (BookedEnd seatBooked)
```

Monad, Carry Thy State!

- One solution would be linear types
- Another is to carry the state *inside* the monad
- No need for explicit state values:

```
payTwice = do
  payForSeat
  payForSeat -- BOOM, type error!
end
```

- We parameterize the monad, or *index* it, by the state type

Indexed Monads

Indexed Monad Type Class

- A monad with two extra type parameters:
 - Input
 - Output
- Can be seen as type before and after the computation
- Type class:

```
class IxApplicative m => IxMonad (m :: k -> k -> * -> *) where  
    ...
```

Monad bind

```
bind :: m a -> (a -> m b) -> m b
```

ibind (simplified)

`bind :: m a -> (a -> m b) -> m b`

`ibind :: m i j a -> (a -> m j k b) -> m i k b`

ibind (simplified)

`bind :: m a -> (a -> m b) -> m b`

`ibind :: m i j a -> (a -> m j k b) -> m i k b`



Specializing ibind

ibind

```
  ::      m i      j      a
-> (a  -> m j      k      b )
->      m i      k      b
```

Specializing ibind (cont.)

```
ibind
  ::      m State1 State2 ()
  -> (() -> m State2 State3 ())
  ->      m State1 State3 ()
```

Indexed Bind Example

```
checkout :: m HasItems NoCard ()
```

```
selectCard :: m NoCard CardSelected ()
```

```
(checkout `ibind` const selectCard) :: m HasItems CardSelected ()
```


- We hide the state *value*
- Only the state type is visible
- We cannot evaluate a computation twice *unless the type permits it*

Composability

- The indexed monad describe *one* state machine
- Hard to compose
- We want *multiple* state machines in a single computation
 - Opening two files, copying from one to another
 - Ticket machine using a card reader and a ticket printer
 - A web server and a database connection
- One solution:
 - A type, mapping from names to states, as the index
 - Named state machines are independent
 - Apply events *by name*

Row Types in PureScript

- PureScript has a *row kind* (think type-level record):

```
(out :: File, in :: Socket)
```

- Can be polymorphic:

```
forall r. (out :: File, in :: Socket | r)
```

- Used as indices for record and effect *types*:

```
Record (out :: File, in :: Socket)
```

```
-- is the same as:
```

```
{ out :: File, in :: Socket }
```

Row Types for State Machines

```
-- Creating `myMachine` in its initial state:
initial
  :: forall r
    . m r (myMachine :: InitialState | r) Unit

-- Transitioning the state of `myMachine`.
someTransition
  :: forall r
    . m (myMachine :: State1 | r) (myMachine :: State2 | r) Unit

-- Deleting `myMachine` when in its terminal state:
end
  :: forall r
    . m (myMachine :: TerminalState | r) r Unit
```

Running Row Type State Machines

```
runIxMachines
  :: forall m
  . Monad m
=> IxMachines m () () a      -- empty rows!
-> m a
```

- `Control.ST` in Idris contrib library⁵
- “purescript-leffe” (The **L**abeled **E**ffects **E**xtension)⁶
- “Motor” for Haskell⁷

⁵ <http://docs.idris-lang.org/en/latest/st/state.html>

⁶ <https://github.com/owickstrom/purescript-leffe>

⁷ <http://hackage.haskell.org/package/motor>

- Read the introduction on “Kwang’s Haskell Blog”⁸
- Haskell package indexed⁹
- Also, see RebindableSyntax language extension
- Can be combined with session types¹⁰

⁸ <https://kseo.github.io/posts/2017-01-12-indexed-monads.html>

⁹ <https://hackage.haskell.org/package/indexed>

¹⁰ Riccardo Pucella and Jesse A. Tov, Haskell session types with (almost) no class, Haskell '08.

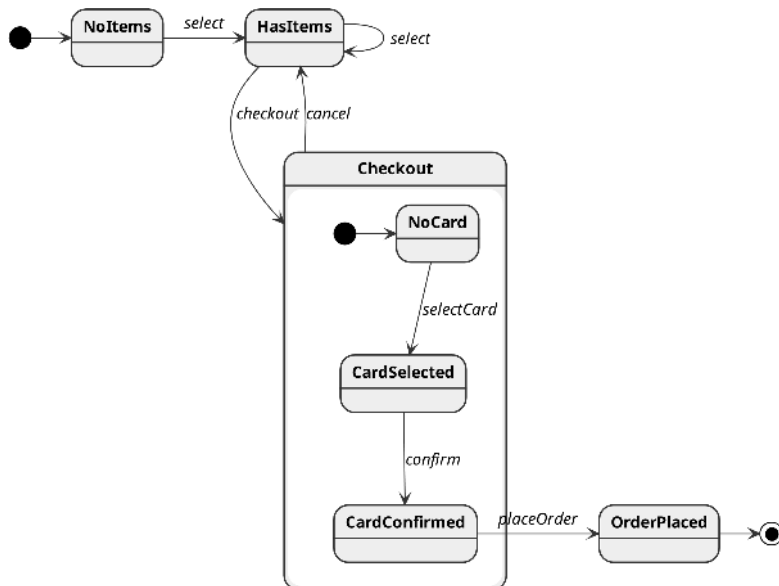
Dependent Types in Idris



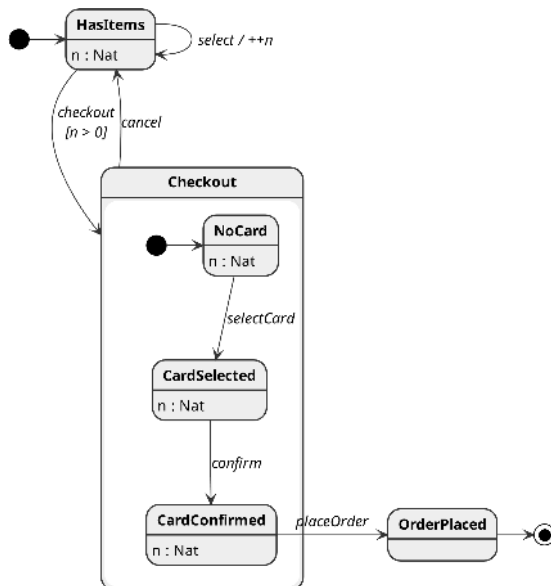
- Dependent types makes some aspects more concise
 - Multiple states accepting an event
 - Error handling
 - Dependent state types
- The `Control.ST` library in Idris supports multiple “named” resources
- “Implementing State-aware Systems in Idris: The ST Tutorial”¹¹

¹¹ <http://docs.idris-lang.org/en/latest/st/index.html>

Revisiting Checkout



Extended State HasItems



Protocol Namespace

```
namespace Protocol
```

```
Item : Type
```

```
Item = String
```

```
Items : Nat -> Type
```

```
Items n = Vect n Item
```

```
Card : Type
```

```
Card = String
```

```
OrderId : Type
```

```
OrderId = String
```

```
...
```

Checkout States

```
data CheckoutState
  = HasItems Nat
  | NoCard Nat
  | CardEntered Nat
  | CardConfirmed Nat
  | OrderPlaced
```

Checkout Interface

```
interface Checkout (m : Type -> Type) where
  State : CheckoutState -> Type

  ...
```

Initial State

```
initial  
  : ST m Var [add (State (HasItems 0))]
```

One More Item

```
select
  : (c : Var)
  -> Item
  -> ST m () [c ::: State (HasItems n)
               :-> State (HasItems (S n))]
```


Checking Out Requires Items

```
checkout
  : (c : Var)
  -> ST m () [c ::: State (HasItems (S n))
              :-> State (NoCard (S n))]
```

States Accepting Cancel

- Again, we have *three* states accepting cancel
- In Idris we can express this using a predicate over states
- “Give me proof that your current state accepts cancel”

Cancellable State Predicate

```
data CancelState : CheckoutState -> (n : Nat) -> Type where  
  
  NoCardCancel : CancelState (NoCard n) n  
  
  CardEnteredCancel : CancelState (CardEntered n) n  
  
  CardConfirmedCancel : CancelState (CardConfirmed n) n
```

Cancelling

```
cancel
  : (c : Var)
  -> { auto prf : CancelState s n }
  -> ST m () [c ::: State s
               :-> State (HasItems n)]
```

Console Checkout Program

```
total
selectMore
  : (c : Var)
  -> ST m () [c ::: State {m} (HasItems n)
              :-> State {m} (HasItems (S n))]]
selectMore c {n} = do
  if n == 0
    then putStrLn "What do you want to add?"
    else putStrLn "What more do you want to add?"
  item <- getStr
  select c item
```

Console Checkout Program (cont.)

```
total
checkoutWithItems
  : (c : Var)
  -> ST m Bool [c ::: State {m} (HasItems (S n))
                :-> (State {m} OrderPlaced
                    `orElse`
                    State {m} (HasItems (S n)))]
checkoutWithItems c = do
  checkout c
  True <- continueOrCancel c | False => pure False
  putStrLn "Enter your card:"
  selectCard c !getStr
  True <- continueOrCancel c | False => pure False
  confirm c
  True <- continueOrCancel c | False => pure False
  placeOrder c
  pure True
```

Console Checkout Program (cont.)

```
total  
checkoutOrShop  
  : (c : Var)  
  -> STLoop m () [remove c (State {m} (HasItems (S n)))]  
checkoutOrShop c = do  
  True <- checkoutWithItems c | False => goShopping c  
  orderId <- end c  
  putStrLn ("Checkout complete with order ID: " ++ orderId)  
  pure ()
```

Console Checkout Program (cont.)

```
total  
goShopping  
  : (c : Var)  
  -> STLoop m () [remove c (State {m} (HasItems n))]  
goShopping c = do  
  selectMore c  
  putStrLn "Checkout? (y/n)"  
  case !getStr of  
    "y" => checkoutOrShop c  
    _   => goShopping c
```


Console Checkout Program (cont.)

total

```
program : STransLoop m () [] (const [])
```

```
program = do
```

```
  c <- initial
```

```
  goShopping c
```

Console Checkout Program (cont.)

```
runCheckout : IO ()  
runCheckout =  
    runLoop forever program (putStrLn "Oops.")
```

Summary

- Implicit state is hard and unsafe when it grows
 - Very unclear, no documentation of states and transitions
 - “Better safe than sorry” checks all over the place
- Just making the states explicit is a win
 - You probably have “hidden” state machines in your code
 - Use data types for states and events (ADTs)
 - This can be done in most mainstream languages!

Summary (cont.)

- By lifting more information to types, we can get more safety
 - You can do *a lot* in Haskell and PureScript
 - Protect side-effects with checked state transitions
 - Even better documentation
 - Make critical code testable
- Steal ideas from other languages
 - Dependent types, linear types
- No silver bullet

Reify your design in code.

Questions?

- Slides and code:

github.com/owickstrom/fsm-your-compiler-wants-in

- Website: <https://wickstrom.tech>
- Twitter: [@owickstrom](https://twitter.com/owickstrom)